

8 Natural selection and artificial selection (selective breeding) both have important roles in evolutionary change.

(a) Seahorses are fish with an unusual appearance and many specialised features.

Fig. 8.1 shows a seahorse.

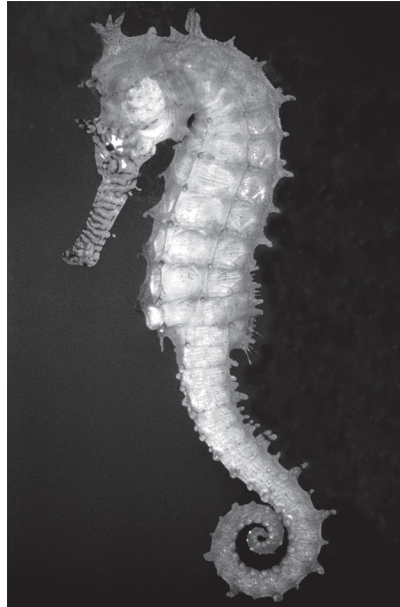


Fig. 8.1

Two species of seahorse, *Hippocampus erectus* and *H. zosterae*, are found in the coastal waters of the Gulf of Mexico and the Caribbean Sea.

- It is thought that these two species of seahorse have a common ancestor.
- The ranges of the two species overlap in many areas.
- Male and female seahorses stay together for the whole of the breeding season.
- *H. erectus* is much larger than *H. zosterae*.
- Few seahorses occur that are intermediate in size between the two species.

Use this information to name **and** explain the type of speciation that may have occurred in the evolution of these two species.

.....

.....

.....

.....

.....

.....

..... [3]

- (b) For many years, farmers have used selective breeding to improve the milk yield of dairy cattle and to improve the yield of crops.

Explain why improving milk yields in cattle by selective breeding can be more challenging than improving yields of crops by selective breeding.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [3]

- (c) State **two** examples of crop features that may be improved by selective breeding to increase the yield of crops.

.....

.....

.....

.....

..... [2]

[Total: 8]